



MATHIA

ALGEBRA

Formule — Kompletan pregled formula

I deo

Relacije, strukture, kompleksni brojevi i polinomi

II deo

Matrice, determinante, vektori, sistemi, prostori

Uči s ljubavlju

Tvoja profesorka

Relacije i funkcije

$$(\mathbf{R}) a \rho a \quad (\mathbf{S}) a \rho b \Rightarrow b \rho a \quad (\mathbf{AS}) (a \rho b \wedge b \rho a) \Rightarrow a = b \quad (\mathbf{T}) (a \rho b \wedge b \rho c) \Rightarrow a \rho c$$

$$(g \circ f)(x) = g(f(x)) \quad (g \circ f)^{-1} = f^{-1} \circ g^{-1}$$

Bulove algebre

$$a(b + c) = ab + ac \quad a + a' = 1, a \cdot a' = 0 \quad (a + b)' = a'b', (ab)' = a' + b'$$

Grupe, prsteni, polja

$$\text{grupa: asoc., neutral, inverz} \quad \text{polje: } \forall a \neq 0 \exists a^{-1} \quad \mathbb{Z}_p \text{ polje} \iff p \text{ prost}$$

Konačna polja

$$GF(p) = \mathbb{Z}_p \quad GF(p^n) = \mathbb{Z}_p[x]/(f), f \text{ ireducibilan, } \deg f = n, |GF(p^n)| = p^n$$

Kompleksni brojevi

$$z = r e^{i\varphi}, r = |z| = \sqrt{x^2 + y^2} \quad z^n = r^n e^{in\varphi} \quad \sqrt[n]{z} = \sqrt[n]{r} e^{i(\varphi + 2k\pi)/n}$$

Polinomi

$$p(\alpha) = 0 \iff (x - \alpha) \mid p(x) \quad p(x) = a_n \prod_k (x - z_k) \quad x_1 + x_2 = -\frac{b}{a}, x_1 x_2 = \frac{c}{a}$$

Matrice i determinante

$$(AB)^T = B^T A^T \quad (AB)^{-1} = B^{-1} A^{-1} \quad \det(AB) = \det A \det B \quad A^{-1} = \frac{1}{\det A} \text{adj}(A)$$

Vektori

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \varphi \quad \vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \quad (\vec{a}, \vec{b}, \vec{c}) = \vec{a} \cdot (\vec{b} \times \vec{c})$$

Analitička geometrija

$$\vec{r} = \vec{r}_P + t\vec{p} \quad \vec{n} \cdot (\vec{r} - \vec{r}_N) = 0 \quad \vec{r}_C = \vec{r}_P + \frac{(\vec{r}_B - \vec{r}_P) \cdot \vec{p}}{\vec{p} \cdot \vec{p}} \vec{p}$$

Sistemi i prostori

$$x_i = \frac{\det A_i}{\det A} \quad \text{rang } A = \text{rang } [A|b] \quad \dim(\text{im } f) + \dim(\ker f) = \dim V$$

Sopstvene vrednosti

$$\det(A - \lambda I) = 0 \quad (A - \lambda I)v = 0, \ v \neq 0$$